H.T.No. | | | | | | | | | | |

Code No: CS4502 **SRGEC-R23**

III B.Tech I Semester Regular Examinations, November 2025

FORMAL LANGUAGES AND AUTOMATA THEORY

(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 70**

Note: All questions from **PART-A** are to be answered at one place.
Answer one question from each unit of **PART-B.** All questions carry equal marks.

PART-A

10 × 2 = 20M

1. a) Construct a transition table for a DFA that accepts all strings over {0,1} that end with '1'.
   b) Write the tuple representation of DFA.
   c) Draw an NFA for the regular expression a*.
   d) Obtain a regular expression to accept strings of a's and b's where number of a's and b's are odd.
   e) Write the leftmost and rightmost derivations of the string aab using the grammar: S→aS | b | a
   f) State Arden's theorem.
   g) Write the transition function equation in tuple format of PDA.
   h) What is the necessity of two-stack PDA?
   i) Give the relationship between turing machine and pushdown automata with respect to stack.
   j) List different types of turing machines.

PART-B

5 × 10 = 50M

UNIT-I

2. Discuss the steps involved in the conversion of NFA into DFA. Construct an equivalent DFA for the NDFA M = ({q1, q2, q3}, {0,1}, δ, q1, q3) where δ is given by: (10M)

   δ(q1,0)={ q2,q3 }, δ(q1,1)={ q1 },

   δ(q2,0)={ q1,q2 }, δ(q2,1)= ∅

   δ(q3,0)={ q2} , δ(q3,1)={ q1,q2 }

(OR)

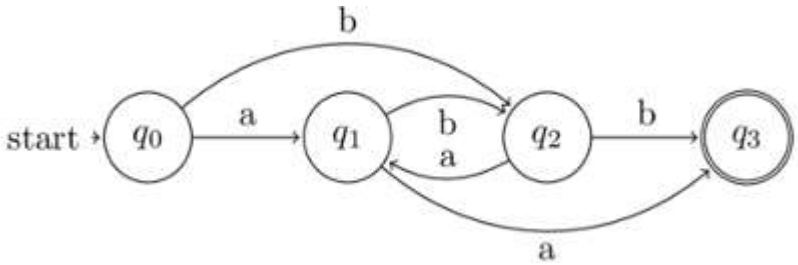
3. a) Draw Transition diagram for the Transition table given below and check acceptance for the strings:
(i) 1000101 (ii) 011001 (6M)

	0	1
→ q0	q1	q0
q1	q2	q0
q2	q3	q0
(q3)	q3	q3

b) Design a DFA to accept strings of a's and b's having even number of a's and b's (4M)

UNIT-II

4. a) Derive a regular expression for the following DFA. (5M)



b) Write the regular expressions for each of the following languages over ∑={0, 1} (5M)

(i) The set of all strings representing with at least two consecutive 0's

(ii) The set of all strings ending in 011

(iii) The set of all strings starting with 101

(OR)

5. a) Write the regular expression for:

(i) Strings that do not contain three consecutive 0's.

(ii) Strings where the number of occurrences of 1's is divisible by 3. (5M)

b) Discuss the differences between regular grammars and context-free grammars. Include appropriate examples to illustrate when a CFG is necessary over a regular grammar. (5M)

UNIT-III

6. a) Consider the grammar ({S, A, B}, {a, b}, P, S) that has the productions:

S→bA | aB

A→ bAA | aS | a

B→aBB | bS | b

Find an equivalent grammar in CNF. (5M)

b) Show that L= $\{a^n b^n c^n \mid n \geq 0\}$ is not a context free language. (5M)

(OR)

7. a) Let G be the grammar (6M)

S→ aB | bA

A→ a | aS | bAA

B → b | bS | aBB.

For the string aaabbabbba, find

(i) Left most derivation

(ii) Right most derivation

(iii) Parse Tree

b) Draw a derivation tree for the string abaaba for the CFG given by G where P = {S→aSa, S→bSb , S→a | b |Є } (4M)

UNIT-IV

8. a) Compare finite automata and PDA. (4M)

b) Design PDA for the language L={ $a^n b^{2n}$ | n>=1}. (6M)

(OR)

9. Describe the equivalence of Context-Free Grammars (CFGs) and Pushdown Automata.. (10M)

UNIT-V

10. a) Give the tuple notation of turing machine. (4M)

b) Design turing machine for the language L = $\{0^n 1^n \mid n \geq 1\}$. (6M)

(OR)

11. Design turing machine for subtraction of two unary numbers. (10M)

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